Unit V:MOBILE TRANSPORT LAYER AND APPLICATION LAYER

1. Mobile Transport Layer

Traditional TCP (Transmission Control Protocol) ensures **reliable, ordered delivery (**sends pkts in sequence and get ack) of data.

Congestion Control Mechanisms

1. Slow Start

- TCP begins with a **small congestion window (cwnd)**.
- It increases **exponentially** (doubles every RTT – Round Trip Time).
- Purpose: Avoid flooding the network at the beginning. ▯ Example:

cwnd = 1 → 2 → 4 → 8 → 16 …

```
cwnd
   |
16|            *
   |         *
   |      *
   |   *
   | *
   | *
   |*
  +------------------------
1  2  3  4  5  RTT
```

```
cwnd
   |
16|        *  *  *  *
   |      *
   |    *
```

```
|   *
| *
|*
+------------------------
```

2. Fast Retransmit

- Triggered when **3 duplicate ACKs** are received
- Retransmits lost packet without waiting for timeout

Ex: when pkts 1, 2,3,4,5 are sent and if pkt 3 is last, then receiver send ACK3 on receipt of next pkts., Sender understand this and resend pkt again.

Sender → 1 → 2 → 3 → 4 → 5

Receiver gets:
1 → ACK 2
2 → ACK 3
4 → ACK 3 (duplicate)
5 → ACK 3 (duplicate)
... another ACK 3

Sender sees 3 duplicate ACKs → retransmits 3

3. Fast Recovery

- Reduces pkts moderately instead of going back to slow start, after retransmit
- Reduces congestion window moderately and continues transmission, **cwnd**

```
|
16|      *
|     *
|   *
| *
|*
|  ↓ (loss detected)
|   \__
```

```
|      \__  (continues linearly)
+------------------------
```

Implications of Mobility

Wireless/mobile environments differ from wired networks:

- Higher **bit error rates**
- Frequent **disconnections (due to handoffs)**
- Variable Bandwidth & Delay
- TCP mistakes errors as congestion → reduces performance

Classical TCP Improvements

1. Indirect TCP (I-TCP) ● Splits connection into:

  - Wired part (fixed host ↔ base station)
  - Wireless part (base station ↔ mobile host)

- Improves performance but breaks end-to-end semantics

2. Snooping (Monitoring) TCP

- Base station "snoops" TCP packets
- Locally retransmits lost packets
- Maintains end-to-end TCP connection

3. Mobile TCP (M-TCP)

- Handles disconnections better
- Freezes sender during disconnection instead of reducing cwnd

Feature	I-TCP	Snooping TCP	M-TCP
Full Name	Indirect TCP	Snooping TCP	Mobile TCP
Basic Idea	Split connection into 2 parts	Base station monitors & retransmits	Freeze sender during disconnection

Connection Type	Not end-to-end	End-to-end maintained	End-to-end maintained
Where Changes Happen	Base station	Base station	Sender + base station
Handles Wireless Errors	Yes (locally)	Yes (local retransmission)	Yes (avoids cwnd reduction)
Handles Disconnections	Poorly	Limited	Very well
Performance	High	High	High (especially mobility cases)
Complexity	Medium	Medium	Higher
Key Advantage	Isolates wireless issues	Transparent improvement	Best for mobility/disconnections
Key Disadvantage	Breaks TCP	Needs packet semantics inspection	More coordination needed

4. Transaction-Oriented TCP (T/TCP)

- Optimized for **short transactions**
- Reduces connection setup overhead

5. TCP over 2.5G/3G Wireless Networks ● Uses enhancements like:

  - Link-layer retransmissions
  - Larger windows
  - Selective acknowledgments (SACK)

2. **Mobile Application Layer**

Wireless environments need lightweight, efficient protocols.

Wireless Application Protocol (WAP): A suite designed for mobile devices with limited (as they are constraints):

- Bandwidth
- Processing power

- Battery

WAP Architecture Layers

1. **Wireless Application Environment (WAE)**
    - Application framework (like web browser)
    - Uses WML (Wireless Markup Language) WMLScript for small screensize and limited bandwidth instead of HTML

Feature	HTML	WML
Designed For	Desktop / modern web	Early mobile devices
Structure	Page-based	Card & deck-based
Complexity	Rich & flexible	Simple & lightweight
Language Base	SGML/HTML	XML
Multimedia Support	High	Very limited
Usage Today	Widely used	Mostly obsolete as 3G, etc are improved in speed, etc., (used with 2G)

Feature	WML (Old)	Modern Web
Technology	WAP + WML	HTML5 + JS
Speed	Slow	Very fast
UI	Very simple	Rich & interactive
Structure	Cards	Full pages
Status	Obsolete	Standard

2. **Wireless Session Protocol (WSP)**
    - Manages sessions (like HTTP but optimized)
    - Supports connection-oriented and connectionless services
3. **Wireless Transaction Protocol (WTP)**
    - Lightweight transaction protocol
    - Provides reliability without full TCP overhead
4. **Wireless Transport Layer Security (WTLS)** ○ Security layer (based on SSL/TLS) ○ Provides:
        - Encryption
        - Authentication
        - Data integrity
5. **Wireless Datagram Protocol (WDP)**
    - Equivalent to UDP
    - Provides a consistent interface over different networks

Wireless Application Environment (WAE)

- Includes:
    - Micro-browser
    - Scripting (WMLScript)
- Designed for small screens and low bandwidth

Mobile Device
↓
WAE (Micro-browser)
↓
WSP
↓
WTP
↓
WTLS
↓
WDP

↓

WAP Gateway

↓

Internet (Web Server)